7.1 Normal vector and Cartesian equation

Key point · Definition and results. A nonzero normal vector to a line is orthogonal to every direction vector of that line. If $\vec{n} = (a, b) \neq (0, 0)$ is normal and $A(x_A, y_A)$ belongs to d, then d has equation $a(x - x_A) + b(y - y_A) = 0$, that is $ax + by + c = 0$. A possible direction vector is $(-b, a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM} \cdot \vec{n} = 0$. In an orthonormal frame, expanding this product gives the equation. Conversely, the set so defined is the line through A orthogonal to n.

Example 1. Through $A(1, 2)$ with normal $(2, -3)$: $2(x - 1) - 3(y - 2) = 0$, so $2x - 3y + 4 = 0$. The vector $(3, 2)$ is a direction vector of the line: scalar product $6 - 6 = 0$.

Watch out. (a, b) is normal, not a direction vector in general. (a, b) must not be zero.

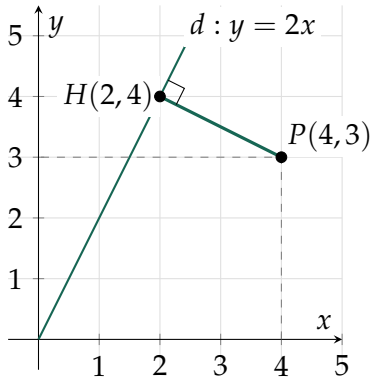
Application. Equation of the line through $(2, -1)$ with normal $(3, 4)$.

7.2 Orthogonal projection and distance

Key point · Definition and results. The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a, b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. The distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a derived extension rather than a standalone routine.

Status and justification. Substitute $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ in the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; we obtain H, then $PH = |\lambda|\sqrt{a^2 + b^2}$.

Example 2. For $P(3, 4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3 + 4 - 2}{2} = \dfrac{5}{2}$, so $H(\dfrac{1}{2}, \dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



The segment PH is perpendicular to d. Both axes have the same scale; the coordinates of P and H can be read directly.

Watch out. The distance is nonnegative: keep the absolute value. Two checks, membership and perpendicularity, verify the projection.

Application. Project $P(4,1)$ onto $d : x = 2$.

7.3 Equation of a circle

Key point · Definition and results. The circle with centre $I(\alpha,\beta)$ and radius $r > 0$ has equation $(x-\alpha)^2+(y-\beta)^2 = r^2$. An equation $x^2+y^2+ux+vy+w = 0$ is handled by completing both squares. If the squared radius obtained is negative, the set is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Since both sides are positive or zero, we can square and use the distance formula in an orthonormal frame.

Example 3. $x^2+y^2-4x+6y+9 = 0$ becomes $(x-2)^2+(y+3)^2 = 4$. Centre $(2,-3)$, radius 2. The signs of the coordinates of the centre are opposite to those appearing in the brackets.

Watch out. A quadratic equation in x and y is not always a circle: check the coefficients and the real radius.

Application. Recognise $x^2+y^2+2x-8y+17 = 0$.

7.4 Solving a configuration

Key point · Definition and results. An intersection is obtained by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel.

Status and justification. The equivalences rest on membership of both sets. The equality $MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Example 4. The circle $x^2+y^2 = 25$ and the line $x = 3$ meet at $(3,4)$ and $(3,-4)$. The perpendicular bisector of $A(0,0)$ and $B(4,2)$ satisfies $x^2+y^2 = (x-4)^2+(y-2)^2$, so $2x+y = 5$.

Watch out. An intersection is not the union of the solution sets. Check the points in both equations.

Application. Intersect $(x-1)^2+(y+2)^2 = 9$ with $y = 1$.